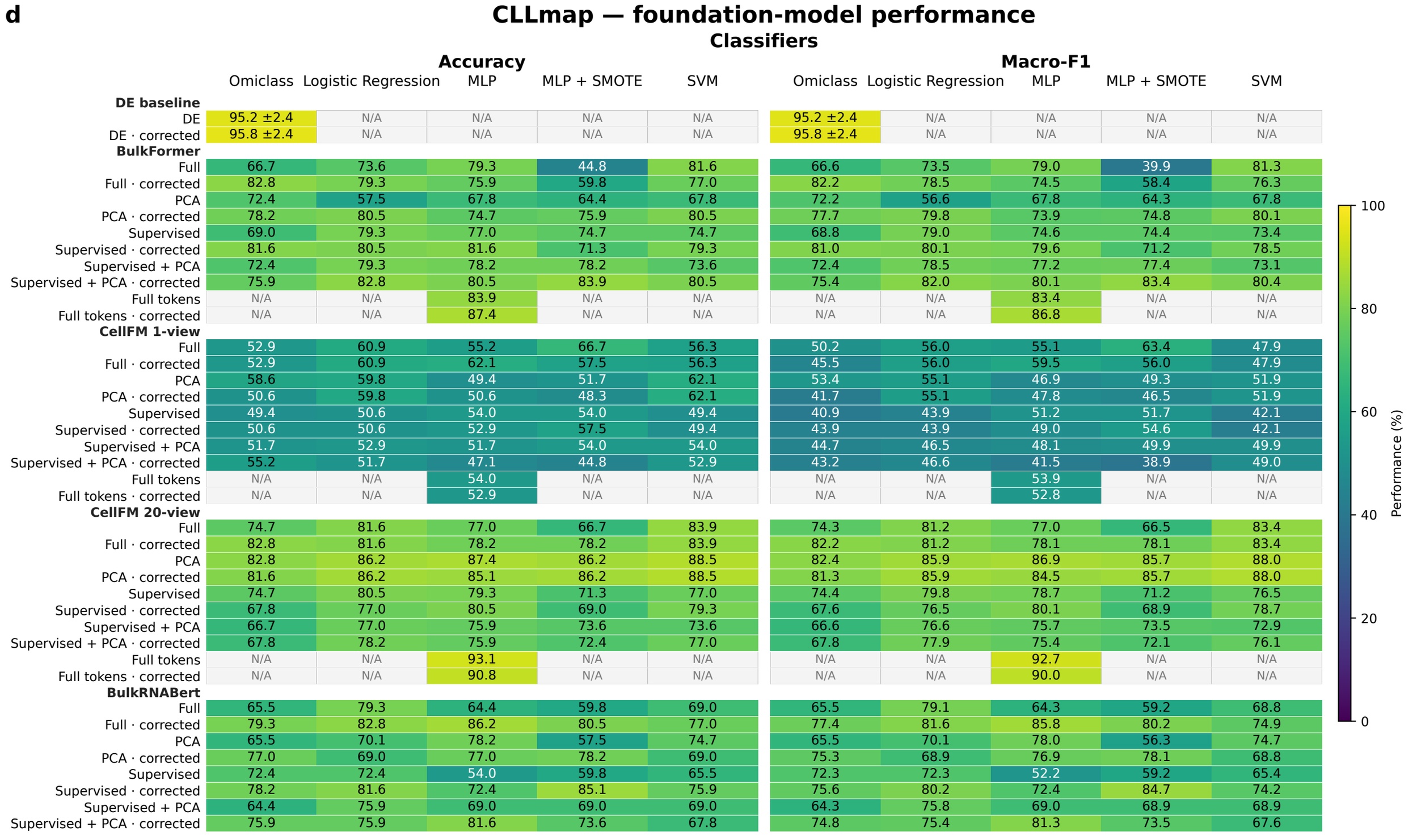
d
CLLmap — foundation-model performance
Classifiers
Accuracy
Macro-F1
Omiclass
Logistic Regression
MLP
MLP + SMOTE
SVM
DE baseline
DE 95.2 ±2.4 N/A N/A N/A N/A 95.2 ±2.4 N/A N/A N/A N/A
DE · corrected 95.8 ±2.4 N/A N/A N/A N/A 95.8 ±2.4 N/A N/A N/A N/A
BulkFormer
Full 66.7 73.6 79.3 44.8 81.6 66.6 73.5 79.0 39.9 81.3
Full · corrected 82.8 79.3 75.9 59.8 77.0 82.2 78.5 74.5 58.4 76.3
PCA 72.4 57.5 67.8 64.4 67.8 72.2 56.6 67.8 64.3 67.8
PCA · corrected 78.2 80.5 74.7 75.9 80.5 77.7 79.8 73.9 74.8 80.1
Supervised 69.0 79.3 77.0 74.7 74.7 68.8 79.0 74.6 74.4 73.4
Supervised · corrected 81.6 80.5 81.6 71.3 79.3 81.0 80.1 79.6 71.2 78.5
Supervised + PCA 72.4 79.3 78.2 78.2 73.6 72.4 78.5 77.2 77.4 73.1
Supervised + PCA · corrected 75.9 82.8 80.5 83.9 80.5 75.4 82.0 80.1 83.4 80.4
Full tokens N/A N/A 83.9 N/A N/A N/A N/A 83.4 N/A N/A
Full tokens · corrected N/A N/A 87.4 N/A N/A N/A N/A 86.8 N/A N/A
CellFM 1-view
Full 52.9 60.9 55.2 66.7 56.3 50.2 56.0 55.1 63.4 47.9
Full · corrected 52.9 60.9 62.1 57.5 56.3 45.5 56.0 59.5 56.0 47.9
PCA 58.6 59.8 49.4 51.7 62.1 53.4 55.1 46.9 49.3 51.9
PCA · corrected 50.6 59.8 50.6 48.3 62.1 41.7 55.1 47.8 46.5 51.9
Supervised 49.4 50.6 54.0 54.0 49.4 40.9 43.9 51.2 51.7 42.1
Supervised · corrected 50.6 50.6 52.9 57.5 49.4 43.9 43.9 49.0 54.6 42.1
Supervised + PCA 51.7 52.9 51.7 54.0 54.0 44.7 46.5 48.1 49.9 49.9
Supervised + PCA · corrected 55.2 51.7 47.1 44.8 52.9 43.2 46.6 41.5 38.9 49.0
Full tokens N/A N/A 54.0 N/A N/A N/A N/A 53.9 N/A N/A
Full tokens · corrected N/A N/A 52.9 N/A N/A N/A N/A 52.8 N/A N/A
CellFM 20-view
Full 74.7 81.6 77.0 66.7 83.9 74.3 81.2 77.0 66.5 83.4
Full · corrected 82.8 81.6 78.2 78.2 83.9 82.2 81.2 78.1 78.1 83.4
PCA 82.8 86.2 87.4 86.2 88.5 82.4 85.9 86.9 85.7 88.0
PCA · corrected 81.6 86.2 85.1 86.2 88.5 81.3 85.9 84.5 85.7 88.0
Supervised 74.7 80.5 79.3 71.3 77.0 74.4 79.8 78.7 71.2 76.5
Supervised · corrected 67.8 77.0 80.5 69.0 79.3 67.6 76.5 80.1 68.9 78.7
Supervised + PCA 66.7 77.0 75.9 73.6 73.6 66.6 76.6 75.7 73.5 72.9
Supervised + PCA · corrected 67.8 78.2 75.9 72.4 77.0 67.8 77.9 75.4 72.1 76.1
Full tokens N/A N/A 93.1 N/A N/A N/A N/A 92.7 N/A N/A
Full tokens · corrected N/A N/A 90.8 N/A N/A N/A N/A 90.0 N/A N/A
BulkRNABert
Full 65.5 79.3 64.4 59.8 69.0 65.5 79.1 64.3 59.2 68.8
Full · corrected 79.3 82.8 86.2 80.5 77.0 77.4 81.6 85.8 80.2 74.9
PCA 65.5 70.1 78.2 57.5 74.7 65.5 70.1 78.0 56.3 74.7
PCA · corrected 77.0 69.0 77.0 78.2 69.0 75.3 68.9 76.9 78.1 68.8
Supervised 72.4 72.4 54.0 59.8 65.5 72.3 72.3 52.2 59.2 65.4
Supervised · corrected 78.2 81.6 72.4 85.1 75.9 75.6 80.2 72.4 84.7 74.2
Supervised + PCA 64.4 75.9 69.0 69.0 69.0 64.3 75.8 69.0 68.9 68.9
Supervised + PCA · corrected 75.9 75.9 81.6 73.6 67.8 74.8 75.4 81.3 73.5 67.6
Performance (%)
100
80
60
40
20
0